4.1 Answer each of the following:

a. Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2.

b. Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.

c. Suppose that a simple regression has quantities $\sum(y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2.

(a) $SST = \sum y_i^2 - N\bar{y}^2 = 7825.94 - 20 \times 19.21^2 = 445.458$

$R^2 = \frac{SSR}{SST} = \frac{375.47}{445.458} \fallingdotseq 0.8429$ ※

(b) $R^2 = 1 - \frac{SSE}{SST} \rightarrow \frac{SSE}{SST} = 1 - R^2$

$SS = SST \times (1 - R^2) = 725.94 \times (1 - 0.7911) = 151.6489$

$\hat{\sigma}^2 = \frac{SSE}{N-2} = \frac{151.6489}{20-2} = 8.4249$ ※

(c) $R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{182.85}{631.63} = 0.7105$ ※

4.3 We have five observations on x and y. They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^{5} \hat{e}_i^2 = 3.6$, $\sum_{i=1}^{5}(x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^{5}(y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

a. the predicted value of y for $x_0 = 4$.

b. the se(f) corresponding to part (a).

c. a 95% prediction interval for y given $x_0 = 4$.

d. a 99% prediction interval for y given $x_0 = 4$.

e. a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

(a)

$x_0 = 4$, $\hat{y}_0 = 1.2 + 0.8 * 4 = 4.4$

(b)

$$\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$$

$$\overline{x} = \frac{\sum x_i}{N} = \frac{3+2+1-1+0}{5} = 1$$

$$se(f) = \sqrt{\hat{var}(f)} = \sqrt{\hat{\sigma}^2[1+\frac{1}{N}+\frac{(x_0-\overline{x})^2}{\sum(x_i-\overline{x})^2}]} = \sqrt{1.2(1+\frac{1}{5}+\frac{3^2}{10})} = 1.5875$$

(c)

A 100(1−α)% prediction interval for y_0 is $\hat{y}_0 \pm t_c se(f)$, where $t_c = t_{(n-2),\frac{\alpha}{2}}$

α = 0.05， df = 5 − 2 = 3， t_c = $t_{0.025,3}$ = 3.1824

$\hat{y}_0 = 4.4$ ，$se(f) = 1.5875$

A 95% prediction interval for y given x_0 = 4 is

[4.4−3.1824*1.5875, 4.4+3.1824*1.5875] = [−0.6520, 9.4520]

(d)

A 100(1−α)% prediction interval for y_0 is $\hat{y}_0 \pm t_c se(f)$, where $t_c = t_{(n-2),\frac{\alpha}{2}}$

α = 0.01， df = 5 − 2 = 3， t_c = $t_{0.005,3}$ = 5.8409

$\hat{y}_0 = 4.4$ ，$se(f) = 1.5875$

A 99% prediction interval for y given x_0 = 4 is

[4.4−5.8409*1.5875, 4.4+5.8409*1.5875] = [−4.8722, 13.6722]

(e)

$$\overline{x} = \frac{\sum x_i}{N} = \frac{3+2+1+(-1)+0}{5} = 1$$

$x_0 = 1$ ，$\hat{y}_0 = 1.2 + 0.8x_0 = 1.2 + 0.8 * 1 = 2$

$$\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{5-2} = 1.2$$

$$\hat{var}(f) = \hat{\sigma}^2[1 + \frac{1}{N} + \frac{(x_0 - \overline{x})^2}{\sum(x_i - \overline{x})^2}] = 1.2 * (1 + \frac{1}{5} + 0) = 1.44$$

$$se(f) = \sqrt{\hat{var}(f)} = \sqrt{1.44} = 1.2$$

A 100(1−α) prediction interval for y_0 is $\hat{y}_0 \pm t_c se(f)$, where $t_c = t_{(n-2),\frac{\alpha}{2}}$

α = 0.05， df = 5 − 2 = 3， $t_c = t_{0.025,3}$ = 3.1824

A 95% prediction interval for y given $x = \bar{x}$ is [2−3.1824*1.2, 2+3.1824*1.2] = [−1.8189, 5.8189]

⇨ The width of part (c) is 9.4520 − (−0.6520) = 10.104,

the width of part (e) is 5.8189 − (−1.8189) = 7.6378.

The prediction interval is narrower for $x = \bar{x}$ compared to x = 4.

This is because predictions are more precise when made for x values close to the mean.

4.25 The file *collegetown* contains data on 500 houses sold in Baton Rouge, LA during 2009–2013. Variable descriptions are in the file *collegetown.def*.

a. Estimate the log-linear model $\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$. Interpret the estimated model parameters. Calculate the slope and elasticity at the sample means, if necessary.

b. Estimate the log-log model $\ln(PRICE) = \alpha_1 + \alpha_2 \ln(SQFT) + e$. Interpret the estimated parameters. Calculate the slope and elasticity at the sample means, if necessary.

c. Compare the R^2 value from the linear model $PRICE = \delta_1 + \delta_2 SQFT + e$ to the "generalized" R^2 measure for the models in (b) and (c).

d. Construct histograms of the least squares residuals from each of the models in (a)–(c) and obtain the Jarque–Bera statistics. Based on your observations, do you consider the distributions of the residuals to be compatible with an assumption of normality?

e. For each of the models in (a)–(c), plot the least squares residuals against $SQFT$. Do you observe any patterns?

f. For each model in (a)–(c), predict the value of a house with 2700 square feet.

g. For each model in (a)–(c), construct a 95% prediction interval for the value of a house with 2700 square feet.

h. Based on your work in this problem, discuss the choice of functional form. Which functional form would you use? Explain.

(a)

```
> summary(m_loglin)

Call:
lm(formula = log(price) ~ sqft, data = collegetown)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.32528 -0.19199  0.00122  0.21678  0.86609 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 4.393866   0.043288  101.50   <2e-16 ***
sqft        0.036044   0.001486   24.26   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.34 on 498 degrees of freedom
Multiple R-squared:  0.5417,	Adjusted R-squared:  0.5408 
F-statistic: 588.7 on 1 and 498 DF,  p-value: < 2.2e-16
```

β_1=4.39， β_2=0.036， $\overline{PRICE} = \overline{y}$ = 250.24， $\overline{SQFT} = \overline{x}$ = 27.28

When $SQFT$ = 0, the value of $ln(PRICE)$ is 4.40. This means that if $SQFT$ = 0, $ln(PRICE)$ has a basic value of 4.40.

When $SQFT$ increase by 1 unit, on average, $PRICE$ will increase by 3.6%

At the sample mean, the slope is $\frac{dy}{dx}$= $\beta_2 \cdot \hat{y} = \beta_2 \cdot \overline{PRICE}$ = 9.02 and the elasticity is $\frac{dy}{dx} \cdot \frac{\overline{x}}{\hat{y}}$ = slope · $\frac{\overline{x}}{\hat{y}}$ = β2·$\overline{SQFT}$ =0.98.

(b)

```
Call:
lm(formula = log(price) ~ log(sqft), data = collegetown)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.36864 -0.20849  0.00487  0.23204  1.21099 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.04965    0.15797   12.97   <2e-16 ***
log(sqft)    1.02483    0.04839   21.18   <2e-16 ***
---
Signif. codes:  
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3643 on 498 degrees of freedom
Multiple R-squared:  0.4738,	Adjusted R-squared:  0.4728 
F-statistic: 448.5 on 1 and 498 DF,  p-value: < 2.2e-16
```

α_1=2.04965

α_2=1.02483

$\overline{PRICE}$=250.24

$\overline{SQFT}$=27.28

α_1: This represents the intercept of the log-log model. It indicates the expected value of ln(PRICE) is 2.04965 when ln(SQFT) is zero.

α_2: This represents the slope of the log-log model. It indicates PRICE will change by 1.025% when SQFT changes by 1%.

At sample mean, slope = 9.3999 , elasticity = 1.0248

An added 100 square feet of interior space increases expected price by $9,400.

(c)

```
Call:
lm(formula = price ~ sqft, data = collegetown)

Residuals:
    Min      1Q  Median      3Q     Max 
-316.93  -58.90   -3.81   47.94  477.05 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -115.4236    13.0882  -8.819   <2e-16 ***
sqft          13.4029     0.4492  29.840   <2e-16 ***
---
Signif. codes:  
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 102.8 on 498 degrees of freedom
Multiple R-squared:  0.6413,	Adjusted R-squared:  0.6406 
F-statistic: 890.4 on 1 and 498 DF,  p-value: < 2.2e-16
```

$modal$	R^2	$generalized\ R^2$
$log - linear$	0.5417	0.6622
$log - log$	0.4738	0.6445
$linear$	0.6413	0.6413

log-linear model: $ln(PRICE) = \beta_1 + \beta_2 SQFT + e$

$\hat{y} = e^{b_1 + b_2 x}$, $R_g^2 = [corr(y, \hat{y})]^2 = r_{y\hat{y}}^2$

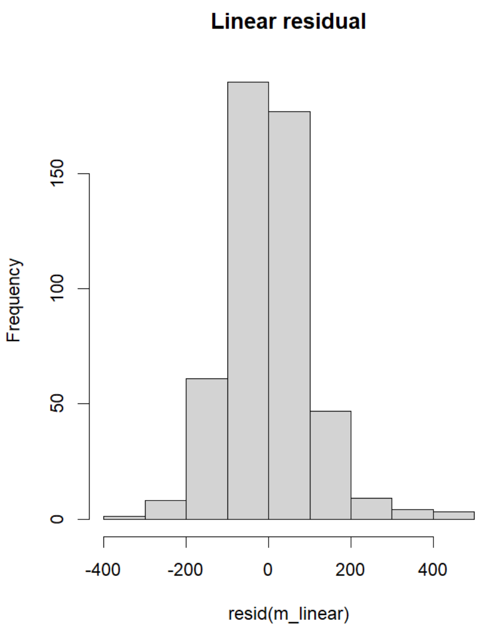
log-log model: $ln(PRICE) = \alpha_1 + \alpha_2 ln(SQFT) + e$

$\hat{y} = e^{a_1 + a_2 ln(x)}$, $R_g^2 = [corr(y, \hat{y})]^2 = r_{y\hat{y}}^2$

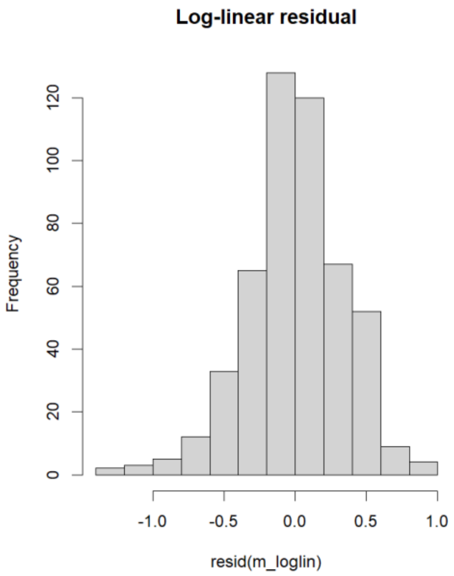
linear model: $PRICE = \delta_1 + \delta_2 SQFT + e$

$\hat{y} = \delta_1 + \delta_2 x$, $R_g^2 = [corr(y, \hat{y})]^2 = r_{y\hat{y}}^2$

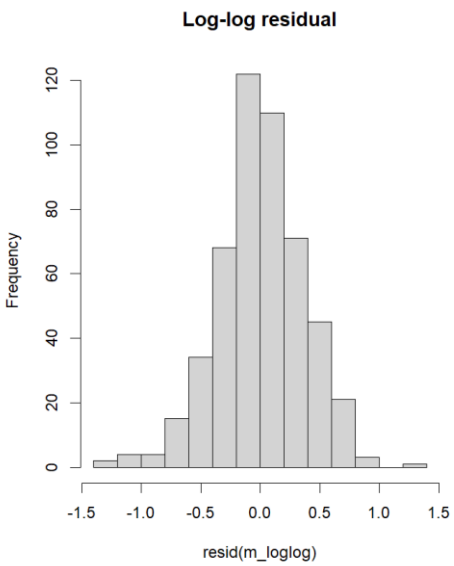
(d)




```
	Jarque Bera Test

data:  resid(m_linear)
X-squared = 221.11, df = 2, p-value < 2.2e-16
```




```
        Jarque Bera Test

data:  resid(m_loglin)
X-squared = 26.679, df = 2, p-value = 1.61e-06
```

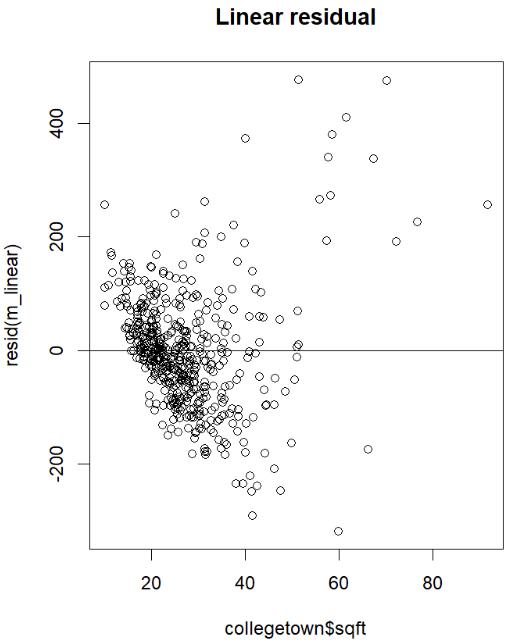


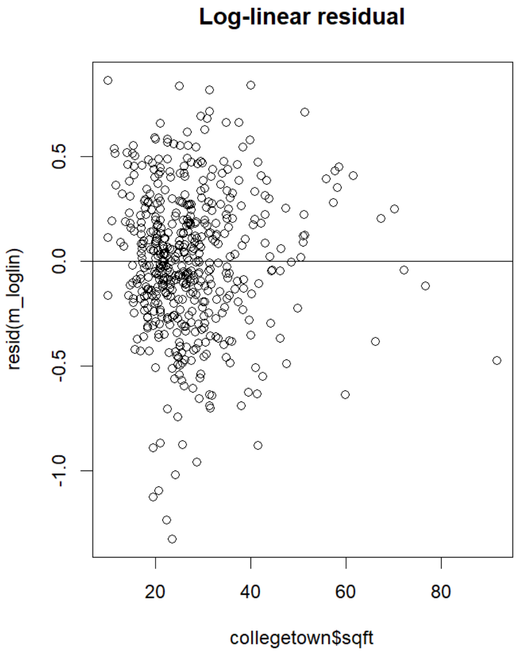

```
        Jarque Bera Test

data:  resid(m_loglog)
X-squared = 14.279, df = 2, p-value = 0.0007933
```

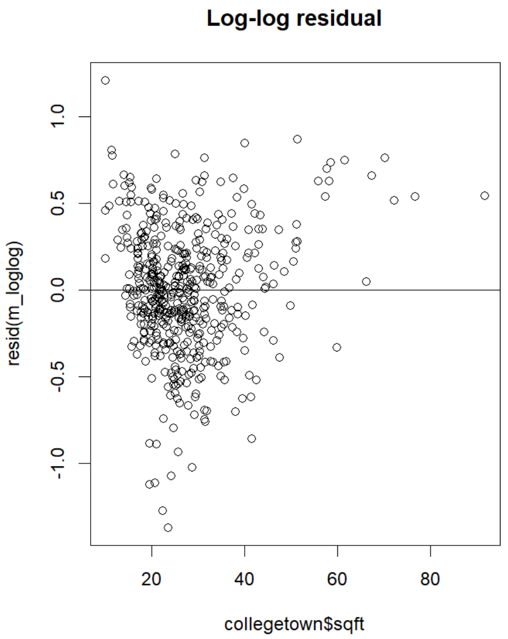
No,the distributions of the residuals do not follow the assumption of normality.

Jarque-Bera test is usually used to test whether the data follows a Normal distribution. The null hypothesis is that the data follows a Normal distribution, and the alternative hypothesis is that the data does not follow a Normal distribution. Since we observe that the Jarque-Bera statistics' P-value is smaller than 5% in all the models, we can conclude that these residuals do not follow the assumption of normality.

(e)







The residual fluctuations of models log-linear and log-log are smaller than linear-linear. Given SQFT, the models of log-linear and log-log satisfy that the expected residual value is 0. However, the linear-linear model does not completely satisfy this condition,which may violate of the homoskedasticity assumption.

(f)

log-linear: $\hat{y} = e^{\beta_1+\beta_2 x}$

log-log: $\hat{y} = e^{\alpha_1+\alpha_2 ln(x)}$

linear: $\hat{y} = \delta_1 + \delta_2 x$

model	log-linear	log-log	linear
predict-value	214,233.6	227,538.6	246,455.7

(Notice that per SQFT stands for 100 sequre feet of house in the data, so we need substitute sqft=2700/100=27. Also per PRICE stands for $1000 in the data, so the final answer should *1000.)

(g)

To calculate the prediction interval , we need to find the unbiased predictor $\hat{y}$ and the standard error of the forecast $se(f) = \sqrt{var(f)}$ and t-statistic $t_{\frac{\alpha}{2}}(n-2)$.

$$var(f) = \sigma^2 \cdot [1 + \frac{1}{N} + \frac{(x_0 - \overline{x}^2)}{\sum(x_i - \overline{x})^2}]$$

The interval for linear model is $[\hat{y} \pm se(f) \cdot t_{\frac{\alpha}{2}}(n-2)]$

The interval for log-linear & log-log model is $[\exp\left(\hat{y} \pm se(f) \cdot t_{\frac{\alpha}{2}}(n-2)\right)]$, in $1000

PI_loglin= [109.769, 418.117]

PI_loglog = [111.141, 466.841]

PI_linear = [44.277, 448.634]

(h)

$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \overline{y})^2}$, $generlized\ R^2 = \left(\text{corr}(y, \hat{y})\right)^2$

$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$, $\text{RMSE} = \sqrt{\text{MSE}}$

	R^2	generalized R^2	MSE	RMSE	jarque's p value	residual
log-linear	0.5417	0.6622	15264.64	123.5501	$1.61e-06$	convergence
log-log	0.4738	0.6445	13190.87	114.8515	0.00079	convergence
linear	0.6413	0.6413	10525.69	102.5948	$< 2.2e-16$	divergence

There is little difference between the log-log and log-linear model results. The log-linear model has a slightly higher generalized R^2 but the log-log model has a slightly smaller Jarque –Bera Statistic. The slopes and elasticities are similar. At this point either model is preferable to the linear relationship.